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Sealing assembly for window air conditioner, and window air conditioner having same

Abstract

A sealing assembly includes a fixation member and a sealing member connected to the fixation member. The fixation member is configured to be connected to a casing of a window air conditioner that is to be supported at a window and has an accommodation groove. The sealing member is configured to be sealingly arranged between a window sash of the window and an inner wall of the window. The window sash is at least partially extended in the accommodation groove.

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATIONS**

(1) The present application is a National Stage Entry under 35 U.S.C. § 371 of International Application No. PCT/CN2020/124109, filed on Oct. 27, 2020, which is based on and claims priority to Chinese Patent Application No. 201922127449.5 filed on Nov. 29, 2019, Chinese Patent Application No. 201911204609.X filed on Nov. 29, 2019, Chinese Patent Application No. 201922500888.6 filed on Dec. 31, 2019, Chinese Patent Application No. 201922496808.4 filed on Dec. 31, 2019, and Chinese Patent Application No. 201922495249.5 filed on Dec. 31, 2019, the entire contents of all of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the field of air conditioners, and more particularly, to a sealing assembly for a window air conditioner and a window air conditioner having same.

BACKGROUND

(3) For a window air conditioner in the related art, a sealing assembly for the window air conditioner has poor sealing performance and a complex structure that is not easy to be mounted, which affect the applicable scope and performance of the window air conditioner.

SUMMARY

(4) The present disclosure aims to solve at least one of the technical problems in the related art.

(5) To this end, the present disclosure provides a sealing assembly for a window air conditioner to improve sealing performance of the sealing assembly.

(6) The present disclosure also provides a window air conditioner having the above sealing assembly.

(7) In the sealing assembly for the window air conditioner according to embodiments of the present disclosure, the window air conditioner is adapted to be supported at a window in a wall body and includes a casing having an accommodation groove. The window has a movable window sash provided therein, and the window sash is adapted to at least partially extend into the

accommodation groove. The sealing assembly is adapted to be in contact with each of the window sash and an inner wall of the window. The sealing assembly includes a fixation member connected to the casing, and a sealing member connected to the fixation member and adapted to be sealingly arranged between the window sash and the inner wall of the window.

(8) With the sealing assembly for the window air conditioner according to the embodiments of the present disclosure, by sealingly arranging the sealing member between the window sash and the inner wall of the window, on the one hand, the sealing assembly has improved sealing performance, and on the other hand, the sealing assembly has a good sound insulation effect. In addition, the sealing member can be cut based on a distance between a side wall surface of the casing and an inner wall surface of the window, such that the sealing member can be mounted more conveniently, which in turn simplifies a structure of the sealing assembly.

(9) In some embodiments of the present disclosure, the fixation member has a mounting groove formed therein. Further, a part of the sealing member is mounted in the mounting groove.

(10) In some embodiments of the present disclosure, the mounting groove has a locking ridge provided on each of opposite side surfaces thereof. Further, the locking ridge is configured to be in contact with the sealing member to position the sealing member within the mounting groove.

(11) In some embodiments of the present disclosure, the window air conditioner further includes a rotation support fixed at the casing. The fixation member is rotatably disposed on the rotation support in such a manner that the sealing assembly is capable of being rotated to be received in the accommodation groove.

(12) In some embodiments of the present disclosure, the rotation support has a rib provided on each of opposite side walls thereof. Further, the rib is configured to be in contact with an inner wall surface of the accommodation groove at a corresponding side.

(13) In some embodiments of the present disclosure, the rib has a guide chamfer provided at a lower end thereof.

(14) In some embodiments of the present disclosure, the rotation support has a sealing plate provided on each of the opposite side walls thereof. Further, the sealing plate extends towards the inner wall surface of the accommodation groove at the corresponding side and is located above the rib on the corresponding side.

(15) In some embodiments of the present disclosure, the sealing assembly further includes an angle positioning assembly configured to cooperate with each of the rotation support and the fixation member to position the fixation member at a predetermined angle when the fixation member is rotated to the predetermined angle.

(16) In a window air conditioner according to embodiments of the present disclosure, the window air conditioner is adapted to be supported at a window in a wall body. The window has a movable window sash provided therein. The window air conditioner includes: a casing having an accommodation groove formed therein, the window sash being adapted to at least partially extend into the accommodation groove; and the sealing assembly according to the embodiments as described above, the sealing assembly being adapted to be sealingly arranged between the window sash and the inner wall of the window.

(17) In the window air conditioner according to the embodiments of the present disclosure, by sealingly arranging the sealing member between the window sash and the inner wall of the window, on the one hand, the sealing assembly has improved sealing performance, and on the other hand, the sealing assembly has a good sound insulation effect. In addition, a sealing member of the sealing assembly can be cut based on a distance between a side wall surface of the casing and an inner wall surface of the window, such that the sealing member can be mounted more conveniently, which in turn simplifies a structure of the sealing assembly.

(18) In some embodiments of the present disclosure, the window air conditioner further includes a positioning device having an unlocked state in which the positioning device is separated from the window sash and a locked state in which the positioning device is brought into contact with the

window sash to position the window sash.

(19) In some embodiments of the present disclosure, the casing includes a base, a rear housing fixed at the base and configured to accommodate an outdoor heat exchanger, a front housing fixed at the base and arranged to be spaced apart from the rear housing to form the accommodation groove, and a middle partition plate fixed at the base and located within the accommodation groove. Further, the middle partition plate is configured to be engaged with the rear housing and the front housing at a front end and a rear end thereof, respectively.

(20) In some embodiments of the present disclosure, the middle partition plate has a placement space provided thereon. Further, the placement space is opened at a top thereof, and the rotation support is received within the placement space. The fixation member has an accommodation space formed therein. An outer edge of the placement space extends, in response to the sealing assembly being rotated out of the accommodation groove, into the accommodation space in such a manner that the sealing assembly is substantially flush with the middle partition plate.

(21) In some embodiments of the present disclosure, the middle partition plate has a drainage hole formed therein. Further, the drainage hole is adapted to drain water in the middle partition plate to an outdoor side.

(22) In some embodiments of the present disclosure, the middle partition plate includes a bottom plate and side plates formed at opposite sides of the bottom plate. A receiving groove is formed between the side plates and the bottom plate, and the drainage hole is arranged at a bottom portion of the receiving groove.

(23) In some embodiments of the present disclosure, the receiving groove of the middle partition plate has a plurality of drainage holes formed therein. Further, at least part of the plurality of drainage holes is adapted to drain water to a part of the base corresponding to the outdoor side. The window air conditioner further includes a water receiving tray mounted on the base. Further, a part of the plurality of drainage holes is adapted to drain water to the water receiving tray to be discharged to an outdoor part of the base through the water receiving tray.

(24) In some embodiments of the present disclosure, the receiving groove is recessed downwardly at each of both ends thereof to form a placement space. The plurality of drainage holes includes a first drainage hole formed at one end of the placement space. Further, the first drainage hole is adapted to drain water to the water receiving tray. The water receiving tray has a drainage groove configured to drain water to the base. The first drainage hole is correspondingly located above the drainage groove. And/or, the plurality of drainage holes include a second drainage hole formed at the other end of the placement space. The second drainage hole is adapted to drain water to the base.

(25) In some embodiments of the present disclosure, the receiving groove is recessed downwardly in a middle portion thereof to form a recess. The plurality of drainage holes further include a third drainage hole formed at a bottom portion of the recess. Further, the third drainage hole is adapted to drain water to the part of the base corresponding to the outdoor side.

(26) In some embodiments of the present disclosure, the recess has a fixation hole passing through a bottom portion thereof. Further, the fixation hole is adapted to be connected and fixed to the water receiving tray by a connection member, and the third drainage hole is located on a periphery of the fixation hole.

(27) In some embodiments of the present disclosure, the water receiving tray partially extends to a position below the middle partition plate and has a support member formed at a part thereof located below the middle partition plate. Further, the support member is adapted to support the middle partition plate.

(28) In some embodiments of the present disclosure, the side plate close to the rear housing has a water overflowing hole formed therein. Further, the water overflowing hole passes through the side plate and is adapted to drain water to the part of the base corresponding to the outdoor side.

(29) In some embodiments of the present disclosure, the base has a support member provided

thereon. Further, the support member corresponds to the accommodation groove, and the middle partition plate is mounted on the support member and adapted to abut with the window sash. The middle partition plate has a positioning member provided on a bottom surface thereof, and the support member is formed with a socket corresponding to the positioning member. Further, the middle partition plate is inserted in and connected to the socket of the support member with the positioning member.

(30) In some embodiments of the present disclosure, the window air conditioner further includes a water receiving tray mounted on the base. Further, the water receiving tray corresponds to the front housing, and the support member is formed on the water receiving tray to be arranged on the base by the water receiving tray, or the support member is formed on the base.

(31) In some embodiments of the present disclosure, a plurality of positioning members are provided. Further, the plurality of positioning members being arranged at intervals in a length direction of the middle partition plate and includes a first positioning member and a second positioning member which are different from each other in shape. A plurality of sockets are provided correspondingly. Further, the plurality of sockets are inserted in and connected to the plurality of positioning members with the plurality of positioning members respectively, and includes a first socket for an insertion of the first positioning member and a second socket for an insertion of the second positioning member.

(32) In some embodiments of the present disclosure, the middle partition plate includes a bottom plate and end side plates formed at both ends of the bottom plate. Further, the bottom plate has the positioning member formed on a bottom surface thereof to be inserted in and connected to the socket of the support member. The end side plates are connected to opposite side walls of the base, respectively.

(33) In some embodiments of the present disclosure, one of the side wall of the base and the end side plate has a laterally protruding snap buckle provided thereon, and the other of the side wall of the base and the end side plate has an engagement hole formed therein. Further, the engagement hole corresponds to the snap buckle and is adapted to be engaged with the snap buckle.

(34) In some embodiments of the present disclosure, each of the end side plates includes a first plate body connected to the bottom plate and a second plate body connected to the first plate body. Further, the second plate body is formed with the snap buckle, and the second plate body and the first plate body being connected to each other at a connection platform configured to abut with an upper edge of the side wall of the base.

(35) In some embodiments of the present disclosure, the middle partition plate further includes side plates formed at opposite sides of the bottom plate. Further, one of the side plates has a support platform formed on an inner surface thereof, and the support platform extends in a length direction of the side plate. The front housing has a notch formed at a back plate thereof. Further, the notch corresponds to the support platform and abuts with the support platform at an upper edge thereof.

(36) In some embodiments of the present disclosure, the middle partition plate further has a fixation part formed at a middle portion thereof. The support member has an engagement member formed thereon. Further, the engagement member corresponds to the fixation part, and the fixation part is connected and fixed to the engagement member by a connection member.

(37) In some embodiments of the present disclosure, the casing has a fixation plate extending therefrom towards the fixation part. The connection member passes through the fixation plate, the fixation part and the engagement member in sequence to integrally connect and fix the middle partition plate to the casing and the support member.

(38) Additional aspects and advantages of the present disclosure will be given at least in part in the following description, or become apparent at least in part from the following description, or can be learned from practicing of the present disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and/or additional aspects and advantages of the present disclosure will become more apparent and more understandable from the following description of embodiments taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a side view of a window air conditioner according to an embodiment of the present disclosure;
- (3) FIG. 2 is a schematic structural perspective view of the window air conditioner illustrated in FIG. 1;
- (4) FIG. 3 is a schematic structural perspective view of the window air conditioner in a state in which a sealing assembly illustrated in FIG. 2 is received in an accommodation groove;
- (5) FIG. 4 is an exploded view of the window air conditioner illustrated in FIG. 1;
- (6) FIG. 5 is an exploded view of a sealing assembly according to an embodiment of the present disclosure;
- (7) FIG. 6 is a top view of the sealing assembly illustrated in FIG. 5;
- (8) FIG. 7 is a schematic structural perspective view of a fixation member illustrated in FIG. 5;
- (9) FIG. 8 is a schematic structural perspective view of a rotation support illustrated in FIG. 5;
- (10) FIG. 9 is a schematic structural perspective view of a window air conditioner having a positioning device according to an embodiment of the present disclosure;
- (11) FIG. 10 is a schematic view showing a window air conditioner mounted on a wall body according to an embodiment of the present disclosure;
- (12) FIG. 11 is schematic structural view of the window air conditioner illustrated in FIG. 10;
- (13) FIG. 12 is a top view of the window air conditioner illustrated in FIG. 10;
- (14) FIG. 13 is a schematic view showing a partial structure of the window air conditioner illustrated in FIG. 12;
- (15) FIG. 14 is a schematic view showing an assembly of a middle partition plate and a base illustrated in FIG. 13;
- (16) FIG. 15 is a schematic structural view of the middle partition plate illustrated in FIG. 14;
- (17) FIG. 16 is a schematic view of the middle partition plate illustrated in FIG. 15 viewed from another perspective;
- (18) FIG. 17 is a schematic structural view of a window air conditioner according to another embodiment of the present disclosure, in which a sealing assembly is in a storage state;
- (19) FIG. 18 a schematic view of the window air conditioner illustrated in FIG. 17 with the sealing assembly switched to an operation state;
- (20) FIG. 19 is a partial schematic view of a window air conditioner according to an embodiment of the present disclosure;
- (21) FIG. 20 is a schematic view of a middle partition plate illustrated in FIG. 19;
- (22) FIG. 21 is another schematic view of the middle partition plate illustrated in FIG. 19;
- (23) FIG. 22 is a perspective view of the window air conditioner illustrated in FIG. 19;
- (24) FIG. 23 is another perspective view of the window air conditioner illustrated in FIG. 22;
- (25) FIG. 24 is an exploded view of a sealing assembly illustrated in FIG. 23;
- (26) FIG. 25 is a top view of a sealing assembly illustrated in FIG. 24;
- (27) FIG. 26 is a schematic structural perspective view of a fixation member illustrated in FIG. 24;
- (28) FIG. 27 is a schematic structural perspective view of a rotation support illustrated in FIG. 24;
- (29) FIG. 28 is a schematic view showing a window air conditioner mounted on a wall body according to an embodiment of the present disclosure;
- (30) FIG. 29 is a schematic view of the window air conditioner illustrated in FIG. 28;
- (31) FIG. 30 is a schematic view showing a partial structure of the window air conditioner

illustrated in FIG. 29;

(32) FIG. 31 is an enlarged view of part A illustrated in FIG. 30;

(33) FIG. 32 is a schematic view showing another partial structure of the window air conditioner illustrated in FIG. 28;

(34) FIG. 33 is an enlarged view of part B illustrated in FIG. 32;

(35) FIG. 34 is a schematic view showing an assembly of a middle partition plate and a base illustrated in FIG. 32;

(36) FIG. 35 is a schematic view showing a disassembly of a middle partition plate and a base illustrated in FIG. 33;

(37) FIG. 36 is an enlarged view of part C illustrated in FIG. 35;

(38) FIG. 37 is a schematic structural view of the middle partition plate illustrated in FIG. 34;

(39) FIG. 38 is a schematic view of a middle partition plate illustrated in FIG. 37 viewed from another perspective;

(40) FIG. 39 is an enlarged view of D illustrated in FIG. 38;

(41) FIG. 40 is an enlarged view of E illustrated in FIG. 38;

(42) FIG. 41 is a schematic structural view of a window air conditioner according to yet another embodiment of the present disclosure, in which a sealing assembly is in a storage state; and

(43) FIG. 42 is a schematic view of a window air conditioner illustrated in FIG. 41 with the sealing assembly switched to an operation state.

DESCRIPTION OF EMBODIMENTS

(44) Embodiments of the present disclosure are described below in detail, examples of the embodiments are shown in accompanying drawings, and throughout the description, the same or similar reference signs represent the same or similar components or the components having the same or similar functions. The embodiments described below with reference to the accompanying drawings are exemplary and merely used to explain the present disclosure, rather than being construed as limitation on the present disclosure.

(45) In the description of the present disclosure, it should be understood that the orientation or position relationship indicated by the terms “center,” “longitudinal,” “transverse,” “length,” “width,” “thickness,” “upper,” “lower,” “front,” “rear,” “left,” “right,” “vertical,” “horizontal,” “top,” “bottom,” “inner,” “outer,” “clockwise,” “counterclockwise,” “axial,” “radial,” “circumferential” etc., is based on the orientation or position relationship shown in the drawings, and is only for the convenience of describing the present disclosure and simplifying the description, rather than indicating or implying that the associated device or element must have a specific orientation or must be constructed and operated in a specific orientation. Thus, the orientation or position relationship indicated by these terms cannot be understood as limitations on the present disclosure. In addition, the features associated with terms “first” and “second” may explicitly or implicitly include at least one of the features. In the description of the present disclosure, “plurality” means at least two, unless otherwise specifically defined.

(46) In the description of the present disclosure, it should be noted that, unless otherwise clearly specified and defined, terms such as “install,” “mount,” “connect to,” “connected with” should be understood in a broad sense. For example, it may be a fixed connection or a detachable connection or integral connection; it may be a mechanical connection or an electrical connection; it may be a direct connection or an indirect connection by an intermediate; it may be an internal communication of two components. For those of ordinary skill in the art, the specific meaning of the above-mentioned terms in the present disclosure can be understood according to specific circumstances.

(47) A sealing assembly **100** for a window air conditioner **200** according to an embodiment of the present disclosure will be described below with reference to FIG. 1 to FIG. 9. The window air conditioner **200** is adapted to be supported at a window **310** in a wall body **300**. The window **310** has a movable window fan **400** provided therein. The window fan **400** is movably arranged within

the window **310** in an up and down direction to open or close the window **310**.

(48) The window air conditioner **200** includes a casing **210** having an accommodation groove **211**, and the window fan **400** is adapted to at least partially extend into the accommodation groove **211**. It should be understood that the casing **210** is divided into an indoor part and an outdoor part by the accommodation groove **211**, and the window fan **400** is at least partially extendable into the accommodation groove **211**. The indoor part has an indoor heat exchanger and an indoor fan that are provided therein, and the outdoor part has an outdoor heat exchanger and an outdoor fan that are provided therein.

(49) Further, the accommodation groove **211** is recessed downwards from a top wall of the casing **210**. Thus, the window air conditioner **200** may not only be exerted with force more uniformly to avoid a damage to the top wall of the window air conditioner **200** due to a great force, thereby improving installation reliability and operation performance of the window air conditioner **200**, but also an air outlet of the window air conditioner **200** may be arranged at a high position to facilitate flowing of an outlet airflow in an indoor space, thereby facilitating an increase in temperature regulation efficiency of the window air conditioner **200** and an improvement of a regulation effect on an indoor temperature of the window air conditioner **200**.

(50) In the sealing assembly **100** for the window air conditioner **200** according to the embodiments of the present disclosure, the sealing assembly **100** is adapted to be in contact with each of the window fan **400** and an inner wall of the window **310**. It should be understood that in a state where the window sash **400** closes the window, the sealing assembly **100** can be brought into contact with the window sash **400** at one side thereof and with the inner wall of the window **310** at the other side thereof, thereby improving the sealing performance of the sealing assembly **100**.

(51) Specifically, the sealing assembly **100** includes a fixation member **1** and a sealing member **2**. The fixation member **1** is connected to the casing **210**. The sealing member **2** is connected to the fixation member **1** and adapted to be sealingly arranged between the window sash **400** and the inner wall of the window **310**. The sealing member **2** may be made of an elastic substance such as a sponge or rubber. It should be understood that the sealing member **2** is connected to the casing **210** by the fixation member **1**. In the state where the window sash **400** closes the window, the sealing member **2** is brought into contact with the window sash **400** at one side thereof and with the inner wall of the window **310** at the other side thereof, and the window is therefore sealed by the sealing member **2**. In this manner, on the one hand, the sealing assembly **100** can have improved sealing performance, and on the other hand, the sealing assembly **100** can have a good sound insulation effect.

(52) Further, the sealing member **2** may have an adjustable length which can be cut on site based on a distance between a side wall surface of the casing **210** and an inner wall surface of the window **310**, such that the sealing member **2** can provide better sealing to the window, thereby ensuring sealing of the window while the sealing assembly **100** has a simpler structure.

(53) In the sealing assembly **100** for the window air conditioner **200** according to the embodiments of the present disclosure, by sealingly arranging the sealing member **2** between the window sash **400** and the inner wall of the window **310**, on the one hand, the sealing assembly **100** has improved sealing performance, and on the other hand, the sealing assembly **100** has a good sound insulation effect. In addition, the sealing member **2** can be cut based on the distance between the side wall surface of the casing **210** and the inner wall surface of the window **310**, such that the sealing member **2** can be mounted more conveniently, which in turn simplifies a structure of the sealing assembly **100**.

(54) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the fixation member **1** has a mounting groove **11** formed therein, and a part of the sealing member **2** is mounted within the mounting groove **11**. That is, by fixing the part of the sealing member **2** in the mounting groove **11**, a connection structure between the sealing member **2** and the fixation member **1** is simpler and more reliable, which in turn allows sealing member **2** to be connected to the casing **210**

easily, and the sealing assembly **100** thus has a simpler structure.

(55) In some embodiments of the present disclosure, the sealing member **2** may be a sealing sponge. For example, the sealing member **2** may be made of a polyvinyl alcohol (PVA) material, so that the sealing assembly **100** has unique strong bonding, film flexibility, smoothness, oil resistance, solvent resistance, protective colloid property, gas barrier property, wear resistance, and water resistance after a special treatment. Therefore, entry of the water into the indoor space is avoided, thereby improving the water resistance of the sealing assembly **100**.

(56) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the mounting groove **11** has a locking ridge **111** provided on each of opposite side surfaces thereof, and the locking ridge **111** is configured to be in contact with the sealing member **2** to position the sealing member **2** within the mounting groove **11**. That is, by arranging the locking ridge **111** on each of the opposite side surfaces of the mounting groove **11**, the locking ridge **111** is brought into contact with the sealing member **2** in the mounting groove **11** to lock the sealing member **2** into the mounting groove **11**, thereby improving a strength of a connection between the sealing member **2** and the fixation member **1**. In this manner, the connection between the sealing member **2** and the fixation member **1** is more stable and reliable.

(57) In some embodiments of the present disclosure, the locking ridge **111** extends obliquely relative to a side wall of the mounting groove **11**. That is, the locking ridge **111** arranged obliquely can lock the sealing member **2** into the mounting groove **11** more stably, thereby improving stability of the sealing member **2**.

(58) In some embodiments of the present disclosure, an included angle between the locking ridge **111** and a corresponding one of the side surfaces of the mounting groove **11** is in a range from 30° to 90°. In this manner, the locking ridge **111** is capable of tightly locking the sealing member **2** into the mounting groove **11** to implement a more secure connection between the sealing member **2** and the fixation member **1**.

(59) In some embodiments of the present disclosure, the fixation member **1** has a first groove **13** formed at an outer side surface thereof. It should be understood that the first groove **13** is arranged on the outer side surface of the fixation member **1** to facilitate a mould opening of the fixation member **1** in such a manner that uniformity of a wall thickness of the fixation member **1** can be ensured to allow a molten adhesive material to flow easily, and therefore, to facilitate fabrication and processing of the fixation member **1**. In addition, it is convenient that the first groove **13** can increase a friction when assembling and disassembling the fixation member **1**, which in turn facilitates the assembly and disassembly of the fixation member **1**.

(60) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the mounting groove **11** has a plurality of locking ridges **111** provided on each side wall of the opposite side surfaces thereof. That is, the mounting groove **11** has two side surfaces that are arranged opposite to each other, each of which is provided with the plurality of locking ridges **111**. Each of the plurality of locking ridges **111** is in contact with the sealing member **2** to better lock the sealing member **2** in the mounting groove **11**, thereby further making the connection between the sealing member **2** and the fixation member **1** more stable and reliable. In some embodiments of the present disclosure, the mounting groove **11** has three spaced-apart locking ridges **111** provided on each side surface thereof, thereby ensuring the strength of the connection between the sealing member **2** and the fixation member **1**.

(61) Specifically, each locking ridge **111** is formed by a portion of the fixation member **1** projecting towards an inside of the mounting groove **11**, thereby providing a simple structure for the fixation member **1**.

(62) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the sealing assembly **100** further includes a rotation support **3** fixed at the casing **210**. The fixation member **1** is rotatably arranged on the rotation support **3** in such a manner that the sealing assembly **100** is capable of being rotated to be received in the accommodation groove **211**. In this way, it is possible

to not only facilitate mounting and arrangement of the fixation member **1**, but also facilitate a rotation of the fixation member **1** relative to the rotation support **3**, such that the sealing assembly **100** is easily stored to reduce a space occupied by the sealing assembly **100**.

(63) Further, the fixation member **1** has a pivoting shaft **10** provided thereon, the rotation support **3** has a pivoting hole **33** formed therein, and the pivoting shaft **10** and the pivoting hole **33** are configured to be rotatably engaged with each other. In this manner, the pivoting shaft **10** and the pivoting hole **33** can be engaged with each other to facilitate a smooth rotation of the fixation member **1** and an improvement of the reliability of the rotation of the fixation member **1**.

(64) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the rotation support **3** has a rib **31** provided on each of opposite side walls thereof, and the rib **31** is configured to be in contact with an inner wall surface of the accommodation groove **211** at a corresponding side thereof. That is, by bringing the rotation support **3** into contact with the inner wall surfaces of the accommodation groove **211** by the ribs **31** on the opposite side walls of the rotation support **3**, a contact area between the rotation support **3** and the inner wall surfaces of the accommodation groove **211** can be reduced to decrease a friction between the rotation support **3** and the inner wall surfaces of the accommodation groove **211**, thereby facilitating the disassembly of the rotation support **3**. In addition, a structural strength of the rotation support **3** can be increased by the ribs **31**.

(65) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, two ribs **31** that are spaced apart from each other are disposed on each of the opposing side walls of the rotation support **3**. Each of the two ribs **31** is brought into contact with an inner wall surface of the accommodation groove **211** at a corresponding side thereof, thereby reducing the contact area between the rotation support **3** and the inner wall surfaces of the accommodation groove **211** while firming the engagement between the rotation support **3** and the accommodation groove **211**. In addition, the structural strength of the rotation support **3** can be increased by the ribs **31**. By arranging the two ribs **31** to be spaced apart from each other, a mould opening of the rotation support **3** is easy, such that the molten adhesive material can flow easily, thereby facilitating fabrication and processing of the rotation support **3**.

(66) Of course, it should be understood that the number of the ribs **31** on each side wall of the rotation support **3** is not limited to two, and may be set based on as desired. For example, three or more ribs **31** may be arranged on each side wall of the rotation support **3**.

(67) In some embodiments of the present disclosure, the rib **31** has a guide chamfer **311** provided at a lower end thereof. The guide chamfer **311** can allow the rotation support **3** to be easily mounted into the accommodation groove **211**, and thus the disassembly of the rotation support **3** is more convenient.

(68) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the rotation support **3** has a sealing plate **32** provided on each of the opposite side walls thereof, and the sealing plate **32** extends towards the inner wall surface of the accommodation groove **211** at the corresponding side thereof and is located above the rib **31** on the corresponding side. It should be understood that, due to the rib **31** between the rotation support **3** and the inner wall surfaces of the accommodation groove **211**, a gap is formed between the rotation support **3** and the inner wall surfaces of the accommodation groove **211**, resulting in an entry of the water into the gap. By arranging the sealing plate **32** on each of the opposite side walls of the rotation support **3** and positioning each sealing plate **32** above the rib **31** on the corresponding side, the gap can be blocked by the sealing plate **32** to prevent the water from entering the gap. Specifically, the sealing plate **32** may be brought into contact with the inner wall surface at the corresponding side of the accommodation groove **211** to effectively prevent the water from entering the gap.

(69) In some embodiments of the present disclosure, the sealing plate **32** is connected to the rib **31** to increase the structural strength of the rotation support **3**.

(70) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the fixation member **1** includes two engagement protrusions **12** that are spaced apart from each other, each of

which is configured to be rotatably engaged with the rotation support **3**. In this way, the pivoting shaft **10** and a positioning protrusion **41** can be easily arranged such that the fixation member **1** can be easily engaged with the rotation support **3** to rotate relative to the rotation support **3**. Therefore, during transportation of the window air conditioner **200**, the fixation member **1** may be rotated to receive the sealing assembly **100** in the accommodation groove **211**, which is conducive for the transportation of the window air conditioner **200**.

(71) In some embodiments of the present disclosure, each engagement protrusion **12** has a hollow member provided thereon. That is, the engagement protrusion **12** has a hollow structure to facilitate a mould opening of the engagement protrusion **12** in such a manner that uniformity of a wall thickness of the engagement protrusion **12** can be ensured to allow the molten adhesive material to flow easily, and therefore, to facilitate the fabrication and processing of the fixation member **1**.

(72) In some embodiments of the present disclosure, each engagement protrusion **12** has a reinforcing rib **121** provided within the hollow member thereof. That is, by arranging the reinforcing rib **121**, a structural strength of the engagement protrusion **12** can be improved, the pivoting shaft **10** can be supported by the reinforcing rib **121** to improve a strength of the pivoting shaft **10**.

(73) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the sealing assembly **100** further includes an angle positioning assembly **4** configured to cooperate with each of the rotation support **3** and the fixation member **1** to position the fixation member **1** at a predetermined angle when the fixation member **1** is rotated to the predetermined angle. In this way, the fixation member **1** can be positioned at a specific angle. For example, an included angle between the fixation member **1** and a horizontal direction may be 90°, 45°, or 30°. Therefore, a rotation angle of the fixation member **1** can be determined by a user as desired, thereby improving use performance of the sealing assembly **100**.

(74) As illustrated in FIG. 5 to FIG. 8, in some embodiments of the present disclosure, the angle positioning assembly **4** includes the positioning protrusion **41** and a plurality of positioning grooves **42**. The positioning protrusion **41** is disposed on the fixation member **1**. The plurality of positioning grooves **42** are arranged on the rotation support **3** in a circular shape. The positioning protrusion **41** can be engaged with the plurality of positioning grooves **42** in a switchable manner when the fixation member **1** is rotated. The positioning protrusion **41** may be engaged with one of the plurality of positioning grooves **42** to position the fixation member **1**. In this way, the rotation angle of the fixation member **1** can be determined by the positioning protrusion **41** and the plurality of positioning grooves **42**, which is convenient to improve reliability and stability of positioning of the fixation member **1**.

(75) Further, the plurality of positioning grooves **42** may be arranged in the circular shape. The positioning protrusion **41** may be engaged with the plurality of positioning grooves **42** in the switchable manner when the fixation member **1** is rotated. The positioning protrusion **41** may be engaged with one of the plurality of positioning grooves **42** to position the fixation member **1**.

(76) For example, a plurality of positioning protrusions **41** are provided and arranged in a ring shape (e.g., a circular ring shape). The plurality of positioning protrusions **41** are configured to be engaged with the plurality of positioning grooves **42** in one-to-one correspondence. In this manner, a more balanced force can be exerted on the angle positioning assembly **4** to facilitate improving a structural strength of the angle positioning assembly **4**, thereby facilitating improving positioning reliability and accuracy of the angle positioning assembly **4**.

(77) A window air conditioner according to embodiments in a second aspect of the present disclosure will be described below with reference to the accompanying drawings.

(78) As illustrated in FIG. 1 to FIG. 9, the window air conditioner **200** is configured to be supported at a window **310** in a wall body **300**. The window **310** has a movable window sash **400** provided therein. The window air conditioner **200** includes a casing **210** and a sealing assembly **100**. The casing **210** has an accommodation groove **211** formed therein. The window sash **400** is

adapted to at least partially extend into the accommodation groove **211**. The sealing assembly **100** is the sealing assembly **100** according to the above embodiments of the present disclosure and is adapted to be sealingly arranged between the window sash **400** and an inner wall of the window **310**.

(79) In the window air conditioner **200** according to the embodiments of the present disclosure, by sealingly arranging the sealing member **2** between the window sash **400** and the inner wall of the window **310**, on the one hand, the sealing assembly **100** has improved sealing performance, and on the other hand, the sealing assembly **100** has a good sound insulation effect. In addition, the sealing member **2** can be cut based on the distance between the side wall surface of the casing **210** and the inner wall surface of the window **310**, such that the sealing member **2** can be mounted more conveniently, which in turn simplifies a structure of the sealing assembly **100**.

(80) As illustrated in FIG. **9**, the window air conditioner **200** further includes a positioning device **500** having an unlocked state in which the positioning device **500** is separated from the window sash **400** and a locked state in which the positioning device **500** is brought into contact with the window sash **400** to position the window sash **400**, thereby facilitating positioning and locking of the window sash **400** to improve safety. In some embodiments of the present disclosure, the positioning device **500** is rotatable to lock or unlock the window sash **400**, such that the positioning device **500** has a simpler and more reliable structure.

(81) In some embodiments of the present disclosure, the casing **210** may be connected to the wall body **300** by a bracket to implement a more secure connection between the window air conditioner **200** and the wall body **300**.

(82) As illustrated in FIG. **1** to FIG. **5**, in some embodiments of the present disclosure, the casing **210** includes a base **212**, a rear housing **213**, a front housing **214**, and a middle partition plate **215**. The rear housing **213** is fixed at the base **212** and configured to accommodate an outdoor heat exchanger therein. The front housing **214** is fixed at the base **212** and arranged to be spaced apart from the rear housing **213** in a forward-and-backward direction to form the accommodation groove **211**. In this way, the accommodation groove **211** is easily formed, such that not only the window air conditioner **200** conveniently cooperate with the window, but also the casing **210** can be processed and fabricated easily with improved aesthetic appearance beautifying.

(83) Further, the middle partition plate **215** is fixed at the base **212** and located within the accommodation groove **211**. The middle partition plate **215** is engaged with the rear housing **213** and the front housing **214** at a front end and a rear end thereof, respectively. In this way, the window sash **400** is easily abutted against the middle partition plate **215** at a lower surface thereof, which facilitates wiring and drainage of the window air conditioner **200**, thereby improve operation reliability of the window air conditioner **200**.

(84) As illustrated in FIG. **1** to FIG. **5**, in some embodiments of the present disclosure, the middle partition plate **215** has a placement space **2150** formed therein, and the placement space **2150** is opened at a top thereof. The rotation support **3** is received within the placement space **2150**. The fixation member **1** has an accommodation space **216** formed therein. An outer edge of the placement space **2150** extends, in response to the sealing assembly **100** being rotated out of the accommodation groove **211**, into the accommodation space **216** in such a manner that the sealing assembly **100** is substantially flush with the middle partition plate **215**. In this way, the sealing assembly **100**, in a state of sealing the window **310**, can be parallel or substantially parallel to the base **212**, such that a height of the sealing assembly, in the state of sealing the window **310**, relative to the window **310** can be lowered, thereby further ensuring a sealing effect.

(85) Some specific embodiments of a window air conditioner **1000** according to the embodiments of the present disclosure will be described below with reference to FIG. **10** to FIG. **18**.

(86) As illustrated in FIG. **10**, the window air conditioner **1000** may be mounted at a window **A201** in a wall body **A20** for cooling or heating an indoor environment. The window air conditioner **1000** includes a base **A100**, a casing body **A200** (excluding the base **A100**), and a middle partition plate

A400. Here, the casing body **A200** is formed on the base **A100** and has an accommodation groove **A230** formed therein. The accommodation groove **A230** is adapted to allow an entrance of a window sash **A30** located at the window **A201** in the wall body **A20**. The middle partition plate **A400** is mounted in the accommodation groove **A230**, and has a drainage hole **R1** (for example, as illustrated in FIG. 15, a first drainage hole **A401** and/or a second drainage hole **A402**, a third drainage hole **A403**, etc.) formed therein. The drainage hole **R1** is adapted to drain water in the middle partition plate **A400** to an outdoor side. For example, the middle partition plate **A400** has a receiving groove **A430** formed therein, and the drainage hole **R1** is adapted to drain water in the receiving groove **A430** to the outdoor side.

(87) Specifically, the casing body **A200** is divided into a front housing **A210** and a rear housing **A220** by the accommodation groove **A230**. The window air conditioner **1000** further includes a compressor, an outdoor heat exchanger, an outdoor fan, an indoor heat exchanger, and an indoor fan. Here, the compressor, the outdoor heat exchanger, and the outdoor fan are mounted within the rear housing **A220**, and the indoor heat exchanger and the indoor fan are mounted within the front housing **A210**.

(88) After the window air conditioner **1000** is mounted at the window **A201** in the wall body **A20**, the rear housing **A220** of the window air conditioner **1000** is located outdoors, and the front housing **A210** of the window air conditioner **1000** is located indoors. In this manner, noises generated by members at the outdoor side can be prevented from propagating to an indoor side, thereby achieving a noise reduction effect. Then, the window sash **A30** is pulled downwardly to extend into the accommodation groove **A230** of the window air conditioner **1000** until a lower edge of the window sash **A30** extends into the middle partition plate **A400**. The window sash **A30** can block a gap between both sides of the window air conditioner **200** and side walls of the window **A201** to avoid a leakage of indoor cold or heated air from the gap to the outdoor space. It should be noted that the window sash **A30** should be understood in a broad sense, and may also be replaced with a window louver, a curtain or the like that blocks light, or other protective window plates that can block an external object from entering the indoor space.

(89) In the above process of mounting the window air conditioner **1000**, compared with a window sash in the related art that can only be pulled up to a top surface of the window air conditioner **1000**, the window sash **A30** according to this embodiment can be inserted into the window air conditioner **1000**. In this way, it is possible for the window sash **A30** to block a space between the both sides of the window air conditioner **1000** and a bottom wall of the window **A201**, thereby increasing an occupation area of the window sash **A30**.

(90) In the related art, since the window air conditioner **1000** is partially exposed to an outdoor environment, water (such as rainwater, condensed water or the like) in the outdoor environment may fall into the accommodation groove **A230** and then overflow into an indoor room from a bottom portion of the accommodation groove **A230**. However, in the window air conditioner **1000** according to the present disclosure, the middle partition plate **A400** is mounted at the bottom portion of the accommodation groove **A230**, and the drainage hole **R1** is formed in the middle partition plate **A400**, such that the water in the outdoor environment may firstly fall onto the middle partition plate **A400** via the accommodation groove **A230**, and then be discharged through the drainage hole **R1** of the middle partition plate **A400** to the outdoor side.

(91) As for a fixing manner for the middle partition plate **A400**, the middle partition plate **A400** may be connected and fixed to the base **A100** or to the casing body **A200**. By taking the middle partition plate **A400** being connected to the base **A100** as an example, the middle partition plate **A400** may be connected to the base **A100** by any one or a combination of an insertion structure, a snapping connection structure, or a screw structure. The present disclosure is not limited in this regard.

(92) In the technical solution of the present disclosure, by mounting the middle partition plate **A400** in the accommodation groove **A230** and forming the drainage hole **R1** in the middle partition plate

A400, the water in the outdoor environment falling into the middle partition plate **A400** can be discharged through the drainage hole **R1** to the outdoor side. Therefore, it can be seen that the window air conditioner **1000** of the present disclosure can decrease an overflowing of the water in the outdoor environment into the indoor room from the accommodation groove **A230**, thereby preventing the indoor room from being wet due to the water.

(93) It is worth mentioning that, as mentioned above, the drainage hole **R1** is adapted to drain water in the middle partition plate **A400** to the outdoor side, in which “the outdoor side” may refer to the outdoor environment or a part of the base **A100** corresponding to the outdoor side. For example, the middle partition plate **A400** has a drainage hole **R1** formed in an end surface thereof, through which the water is discharged to the outdoor environment directly from a side of the window air conditioner **1000** by a drainage pipe. Alternatively, the middle partition plate **A400** has a drainage hole **R1** formed at the bottom portion thereof to drain water to a part of the base **100** corresponding to the outdoor side. Then, the water can be discharged from the base **A100** to the outdoor environment, or spattered to the outdoor heat exchange by a water spattering wheel for dissipating heat of the outdoor heat exchanger.

(94) In some embodiments of the present disclosure, as illustrated in FIG. 13 to FIG. 15, the middle partition plate **A400** includes a bottom plate **A410** and side plates **A420** formed at opposite sides of the bottom plate. The receiving groove **A430** is formed between the side plates **A420** and the bottom plate **A410**, and the drainage hole **R1** is adapted to drain water in the receiving groove **430** to the outdoor side. The receiving groove **A430** of the middle partition plate **A400** has a plurality of drainage holes **R1** formed therein. At least a part of the plurality of drainage holes **R1** is adapted to drain water to the part of the base **A100** corresponding to the outdoor side. The plurality of drainage holes **R1** are arranged at intervals in a length direction of the middle partition plate **A400**. The window air conditioner **1000** further includes a water receiving tray **A600** mounted on the base **A100**. A part of the plurality of drainage holes **R1** is adapted to drain water to the water receiving tray **A600**, and then the water is discharged from the water receiving tray **A600** to an outdoor part of the base **A100**.

(95) Specifically, it should be understood that a part of the plurality of drainage holes **R1** on the middle partition plate **A400** is configured to drain water directly to the part of the base **A100** corresponding to the outdoor side, while another part of the plurality of drainage holes **R1** is configured to drain water firstly to the water receiving tray **A600**, the water is then discharged from the water receiving tray **A600** to the base **A100**.

(96) As illustrated in FIG. 14 to FIG. 16, it is taken into consideration here that when the receiving groove **A430** of the middle partition plate **A400** contains a small amount of water, the water in the receiving groove **A430** is dispersed without easily flowing into the drainage hole. To solve this problem, optionally, the receiving groove **A430** is recessed downwardly at each of both ends thereof to form a placement space **A440**. The plurality of drainage holes **R1** includes a first drainage hole **A401** formed at one end of the placement space **A440**, and the first drainage hole **A401** is adapted to drain water to the water receiving tray **A600**; and/or the plurality of drainage holes **R1** includes a second drainage hole **A402** formed at the other end of the placement space **A440**, and the second drainage hole **A402** is adapted to drain water to the base **A100**.

(97) Specifically, the middle partition plate **A400** has the placement space **A440** formed at each of both ends of the receiving groove **A430**. The placement space **A440** is close to the front housing **A210** at one end thereof and close to the rear housing **A220** at the other end thereof. Therefore, the first drainage hole **A401** is formed at the end of the placement space **A440** close to the front housing **A210**, and located above the water receiving tray **A600** to drain water to the water receiving tray **A600**. The second drainage hole **A402** is formed at the end of the placement space **A440** close to the rear housing **A220**, and located above the part of the base **A100** corresponding to the outdoor side to drain water to the part of the base **A100** corresponding to the outdoor side.

(98) Further, in order to facilitate a flow of the water from the water receiving tray **A600** to the base

A100, as illustrated in FIG. 14, the water receiving tray **A600** has a drainage groove **A610** configured to drain water to the base **A100**, and the first drainage hole **A401** is correspondingly located above the drainage groove **A610**. Therefore, when the water discharged from the first drainage hole **A401** happens to fall into the drainage groove **A610** of the water receiving tray **A600** and then is discharged directly from the drainage groove **A610** to the base **A100**, a drainage path is relatively short, which facilitates enhancing drainage efficiency.

(99) Specifically, the water receiving tray **A600** partially extends to a position below the middle partition plate **A400** and has a drainage groove **A610** at a part thereof located below the middle partition plate **A400**. In order to facilitate mounting of the middle partition plate **A400**, the part of the water receiving tray **A600** located below the middle partition plate **A400** also has a support member **A500**. The support member **A500** is adapted to support the middle partition plate **A400**. The support member **A500** and the water receiving tray **A600** are formed into one piece. The middle partition plate **A400** may be connected to the support member **A500** by inserting the middle partition plate **A400** into the support member **A500**, or through a screw structure, and the present disclosure is not limited in this regard.

(100) With continued reference to FIG. 14 to FIG. 16, it is also taken into consideration here that when the receiving groove **A430** of the middle partition plate **A400** contains a large amount of water, the water may submerge the placement space **A440** and possibly overflow from an upper peripheral edge of the receiving groove **A430**. To solve this technical problem, the receiving groove **A430** is recessed downwardly in a middle portion thereof to form a recess **A450**. The plurality of drainage holes **R1** further includes a third drainage hole **A403** formed at a bottom portion of the recess **A450** and adapted to drain water to the part of the base **A100** corresponding to the outdoor side. Therefore, after the water submerges the placement space **A440** of the middle partition plate **A400**, the water can also be discharged from the third drainage hole **A403** to the base **A100**, thereby effectively increasing a drainage amount of the middle partition plate **A400** and improving a drainage rate.

(101) Since the receiving groove **A430** is recessed downwardly in the middle portion thereof to form the recess **A450**, a stress of the bottom plate **A410** of the middle partition plate **A400** is dispersed at a position where the recess **A450** is located, such that the bottom plate **A410** has a high strength and less prone to fracture at this position. Therefore, the recess **A450** may have a fixation hole **A451** passing through a bottom portion thereof. The fixation hole **A451** is adapted to be connected and fixed to the water receiving tray **A600** by a connection member. The connection member may be a screw or a pin-type structure. On this basis, the third drainage hole **A403** is located on a periphery of the fixation hole **A451** to avoid the third drainage hole **A403** from being too close to the fixation hole **A451** to reduce a strength of a plate surface around the fixation hole **A451**.

(102) In some embodiments, as illustrated in FIG. 14 to FIG. 16, it is further taken into consideration that when the receiving groove **A430** of the middle partition plate **A400** contains a large amount of water that submerges the recess **A450**, individual drainage ports cannot drain such a large amount of water timely. Therefore, a side plate **A420** of the middle partition plate **A400** close to the rear housing **A220** has a water overflowing hole **A404** formed therein. The water overflowing hole **A404** passes through the side plate **A420** and is adapted to drain water to the part of the base **A100** corresponding to the outdoor side.

(103) Specifically, one or more water overflowing holes **A404** may be provided. A plurality of water overflowing holes **A404** may be arranged at intervals in a length direction of the side plate **A420**. When a water level in the receiving groove **A430** rises to reach the water overflowing hole **A404**, the water may be discharged from the water overflowing hole **A404** towards the part of the base **A100** corresponding to the outdoor side, thereby further increasing the drainage amount of the middle partition plate **A400** and improving the drainage rate.

(104) As illustrated in FIG. 17 and FIG. 18, based on any one of the above embodiments, the

window air conditioner **1000** further includes a sealing assembly **A300**. The sealing assembly **A300** is movably mounted on the middle partition plate **A400**, and is adapted to be switchable between a storage state and an operation state through a movement. Here, in the storage state, the sealing assembly **A300** is received in the accommodation groove **A230**, and in the operation state, the sealing assembly **A300** extends laterally from the accommodation groove **A230** and is adapted to be abutted against the window sash **A30** and/or an inner wall of the window **A201**.

(105) After the window air conditioner **1000** is mounted at the window **A201**, the sealing assembly **A300** is moved to the operation state in such a manner that the sealing assembly **300** extends laterally from the accommodation groove **A230** of the window air conditioner **1000** with a bottom surface of the sealing assembly **A300** abutted against the bottom wall of the window **A201**. Then, the window sash **A30** is pulled downwardly to extend into the accommodation groove **A230** of the window air conditioner **1000** until the lower edge of the window sash **A30** is brought into contact and abuts with each of the middle partition plate **A400** and the sealing assembly **A300**. The sealing assembly **A300** is filed in a gap between the window sash **A30** and the bottom wall of the window **A201** and seals the gap to reduce an amount of cold or heated air that leaks from the indoor space to the outdoor space through the gap. When the window air conditioner **1000** is not needed or is transported, the sealing assembly **300** is moved to the storage state to reduce a space occupied by the sealing assembly **300**, thereby facilitating storage or packaging of the window air conditioner **1000**.

(106) Movable mounting for the sealing assembly **300** may be implemented in various ways. For example, the sealing assembly **300** may be slidably mounted on the middle partition plate **A400**. Alternatively, the sealing assembly **300** may be rotatably mounted on the middle partition plate **A400**. Additionally, alternatively, the sealing assembly **300** may be mounted on the middle partition plate **A400** in a resiliently retractable manner. Specifically, the sealing assembly **300** herein is rotatably connected to the middle partition plate **A400** to allow the sealing assembly **300** to be switchable between the operation state and the storage state through a rotation. Specifically, the sealing assembly **300** is rotatably mounted in the placement space **A440** (as illustrated in FIG. 15) at one end thereof in such a manner that the sealing assembly **300** is rotatable relative to the middle partition plate **A400** and switchable between the storage state and the operation state through the rotation.

(107) Some specific embodiments of a window air conditioner **2000** according to the embodiments of the present disclosure will be described below with reference to FIG. 19 to FIG. 27.

(108) As illustrated in FIG. 19, the window air conditioner **2000** is adapted to be supported at a window in a wall body. The window has a movable window sash provided therein. The window sash is movably disposed within the window in an upward-downward direction to open or close the window. The window air conditioner **2000** further includes a base assembly **B500**, a rear housing **B300**, and a front housing **B200**. Both the rear housing **B300** and the front housing **B200** are arranged on the base assembly **B500**. A middle partition plate **B100** is connected between the rear housing **B300** and the front housing **B200**, and may be located below the window sash. The front housing **B200** is located at an inner side of the window. The rear housing **B300** is located at an outer side of the window. The middle partition plate **B100** may be connected to the front housing **B200** and the rear housing **B300** at both ends in a width direction thereof (e.g., an inward-outward direction illustrated in FIG. 19), respectively.

(109) The middle partition plate **B100** includes a partition plate body **B1**. A recess **B11** opened at a top thereof is formed in the partition plate body **B1** and recessed downwardly. The recess **B11** has a third drainage hole **B111** formed therein and a connection hole **B112** that is configured to be connected to an outer housing **300** of the window air conditioner **2000**. The third drainage hole **B111** is in communication with the base assembly **B500**.

(110) Referring to FIG. 19 and FIG. 20, the recess **B11** may be formed by recessing a part of a surface of the partition plate body **B1** downwardly. The recess **B11** may be arranged adjacent to the

rear housing **B300**. Both the connection hole **B112** and the third drainage hole **B111** pass through a bottom wall of the recess **B11**. Specifically, the window air conditioner **2000** may have a mounting lug **B301** provided on the rear housing **B300** thereof. The mounting lug **B301** has a mating hole **B302** formed therein. The mating hole **B302** cooperates with the connection hole **B112**. The mounting lug **B301** may be attached at a position below the recess **11**. Therefore, a fixation piece may pass through and received within the connection hole **B112** and the mating hole **B302**, such that the middle partition plate **B100** can be conveniently and firmly fixed to the rear housing **B300**, thereby improving reliability and stability of a connection between the middle partition plate **B100** and the rear housing **B300** as well as assembly efficiency.

(111) Also, rainwater and the like flowing down from the window sash when it rains may flow to the recess **B11** in the partition plate body **B1** and flow to the base assembly **B500** through the third drainage hole **B111**, thereby preventing the rainwater flowing down from the window sash from entering the indoor space, which in turn prevents a floor and articles in the indoor space from getting wet and damaged. Therefore, safety and reliability of the window air conditioner **2000** can be improved.

(112) It should be understood that water discharged from the third drainage hole **B111** may flow to the base assembly **B500**, and be discharged through the base assembly **B500** to the outdoor space when the water in the base assembly **B500** has gathered to a certain amount. Alternatively, the water discharged from the third drainage hole **B111** may be directly discharged through the base assembly **B500** to the outdoor space.

(113) In some embodiments of the present disclosure, the base assembly **B500** includes a base and a water receiving tray. The water receiving tray is disposed on the base and located below the middle partition plate **B100**. Here, the third drainage hole **B111** may be in communication with the water receiving tray. In this manner, the water discharged from the third drainage hole **B111** can flow into the water receiving tray, thereby avoiding dampness of other members on the base caused by the water discharged from the third drainage hole **B111**, and improving the safety and reliability of the window air conditioner **2000**.

(114) Therefore, by arranging the recess **B11** in the partition plate body **B1** and forming the connection hole **B112** and the third drainage hole **B111** in a mounting slot, the middle partition plate **B100** can be conveniently and firmly fixed to the rear housing **B300**, thereby enhancing the reliability and stability of the connection between the middle partition plate **B100** and the rear housing **B300**, and increasing the assembly efficiency. In addition, the rainwater and the like flowing down from the window sash when it rains may flow to the recess **B11** in the partition plate body **B1** and be discharged to the base assembly **B500** through the third drainage hole **B111**, thereby preventing the rainwater flowing down from the window sash from entering the indoor space, which in turn prevents the floor and articles in the indoor space from getting wet and damaged. Therefore, safety and reliability of the window air conditioner **2000** can be improved.

(115) According to some embodiments of the present disclosure, the partition plate body **B1** has a placement space **B12** formed in each of two ends in a length direction thereof. Referring to FIG. **20** and FIG. **21**, the placement space **B12** may be formed by recessing a part of the surface of the partition plate body **B1** downwardly. The first drainage hole **B121** is formed in the placement space **B12** with in communication with the base assembly **B500**. The first drainage hole **B121** may pass through a bottom wall of the placement space **B12** to drain water from the placement space **B12** timely.

(116) It should be understood that the water discharged from the first drainage hole **B121** may flow into the base assembly **B500**, and be discharged through the base assembly **B500** to the outdoor space when the water in the base assembly **B500** has gathered to a certain amount. Alternatively, the water discharged from the first drainage hole **B121** may be directly discharged through the base assembly **B500** to the outdoor space. Therefore, by arranging the first drainage hole **B121** in the placement space **B12**, discharging water gathered on the partition plate body **B1** to the outdoor

space can be speeded up, such that the gathering of water on the partition plate body B1 can be avoided, thereby preventing the water on the partition plate body B1 from flowing into the indoor space.

(117) In some embodiments of the present disclosure, the base assembly B500 includes a base and a water receiving tray. The water receiving tray is disposed on the base and located below the middle partition plate B100. Here, the first drainage hole B121 may be in communication with the water receiving tray. In this manner, dampness of other members on the base caused by the water discharged from the first drainage hole B121 can be avoided, thereby improving the safety and reliability of the window air conditioner 2000.

(118) Here, the placement space B12 may be configured to mount a sealing assembly B400 of the window air conditioner 2000. The sealing assembly B400 is adapted to be in contact with the window sash and the inner wall of the window, respectively. It should be understood that in a state of the window sash closing the window, the sealing assembly B400 is brought into contact with the window sash at one side thereof and with the inner wall of the window at the other side thereof, thereby improving sealing performance of the sealing assembly B400.

(119) According to some embodiments of the present disclosure, the middle partition plate B100 further includes two end side plates B2. The two end side plates B2 are connected to both ends of the partition plate body B1 in a length direction (e.g., a leftward-rightward direction illustrated in FIG. 19), respectively. Each of the two end side plates B2 has a snap buckle B21 provided thereon. The snap buckle B21 is connected to the base assembly B500 of the window air conditioner 200. For example, referring to FIG. 19 and FIG. 20, the two end side plates B2 may be connected to a left side and a right side of the partition plate body B1, respectively, with one end of each end side plate B2 being connected to the partition plate body B1 and the other end extending downwardly. The base assembly B500 has a buckle slot engaged with the snap buckle B21. Therefore, the middle partition plate B100 can be easily connected to the base assembly B500 to avoid a movement of the middle partition plate B100, thereby improving position stability of the middle partition plate B100.

(120) According to some embodiments of the present disclosure, the base assembly B500 has a socket provided thereon, and the partition plate body B1 has a positioning member B13 provided on a bottom surface thereof. The positioning member B13 is adapted to be engaged with the socket. During assembling, positioning of the middle partition plate B100 can be implemented through the engagement between the socket and the positioning member B13, which effectively lowers a difficulty of assembling the middle partition plate B100 and improves assembly efficiency of the middle partition plate B100.

(121) For example, in some specific embodiments of the present disclosure, the positioning member B13 may be a positioning bump formed on a bottom surface of the partition plate body B1 or a positioning groove formed in the base assembly B500. Here, when the base assembly B500 includes the base and the water receiving tray, the socket may be formed on the water receiving tray. Therefore, the positioning of the middle partition plate B100 can be implemented through an engagement between the positioning bump and the positioning groove. Such a structure is simple and can implement positioning easily.

(122) According to some embodiments of the present disclosure, the positioning member B13 includes a first positioning member B131 and a second positioning member B132 that are spaced apart from each other and have different outer contour shapes. In this manner, during assembling, only when the first positioning member B131 is engaged with the socket corresponding to the first positioning member B131 and the second positioning member B132 is engaged with the socket corresponding to the second positioning member B132, the middle partition plate B100 can be assembled on the base assembly B500. Therefore, fool proofing can be implemented by the first positioning member B131 and the second positioning member B132 that have different outer contours to prevent the middle partition plate B100 from being mounted in a reverse direction,

which is conducive to improving assembly efficiency.

(123) According to some embodiments of the present disclosure, as illustrated in FIG. 21, the first positioning member **B131** is formed into a square bump, and the second positioning member **B132** is formed into a trapezoidal bump. Such a structure is simple and easy to be processed. Of course, it should be understood that the first positioning member **B131** and the second positioning member **B132** may also be formed into other shapes such as a triangle, a circle, an irregular shape, etc., as long as the outer contour shape of the first positioning member **B131** is different from that of the second positioning member **B132**.

(124) According to some embodiments of the present disclosure, as illustrated in FIG. 20 and FIG. 21, the middle partition plate **B100** also includes two side plates **B3** that are connected to both ends of the partition plate body **B1** in a width direction, respectively. A receiving groove **B31** is formed between the two side plates **B3** and the partition plate body **B1**. The receiving groove **B31** is configured to store the rainwater flowing down from the window sash. When a large amount of rainwater flows down from the window sash rapidly, the rainwater can be temporarily stored in the receiving groove **B31** to prevent, when too much water gathered in the recess **B11** cannot be discharged timely, the water from flowing into the indoor space, thereby improving safety and reliability of the window air conditioner **2000**.

(125) According to some embodiments of the present disclosure, the middle partition plate **B100** is made of plastic. Plastic facilitates mass production due to its light weight and low cost, thereby enhancing fabrication efficiency and lowering the cost of the middle partition plate **B100**.

(126) As illustrated in FIG. 22 and FIG. 23, the window air conditioner **2000** includes the base assembly **B500**, the rear housing **B300**, the front housing **B200**, the middle partition plate **B100**, and the fixation piece. The rear housing **B300** is arranged on the base assembly **B500**. The front housing **B200** is arranged on the base assembly **B500**. The front housing **B200** and the rear housing **B300** are spaced apart from each other to form an accommodation groove **B600**. Here, the window sash may be arranged in the accommodation groove **B600**. The middle partition plate **B100** is arranged on the base assembly **B500** and located between the rear housing **B300** and the front housing **B200**. The rear housing **B300** has the mounting lug **B301** provided thereon. The mounting lug **B301** has the mating hole **B302** formed therein. The mating hole **B302** corresponds to the connection hole **B112**. The fixation piece may pass through the connection hole **B112** and the mating hole **B302**.

(127) The mounting lug **B301** may be attached at a position below the recess **B11**. Therefore, the fixation piece may pass through and received within the connection hole **B112** and the mating hole **B302**, such that the middle partition plate **B100** can be conveniently and firmly fixed to the rear housing **B300**, thereby improving reliability and stability of the connection between the middle partition plate **B100** and the rear housing **B300** as well as the assembly efficiency. In addition, the rainwater and the like flowing down from the window sash when it rains may flow to the recess **B11** in the partition plate body **B1** and be discharged through the third drainage hole **B111**, thereby preventing the rainwater flowing down from the window sash from entering the indoor side, which in turn improves the safety and reliability of the window air conditioner **2000**.

(128) According to some embodiments of the present disclosure, the window air conditioner **2000** is adapted to be supported at the window in the wall body. The window has the movable window sash provided therein. The partition plate body **B1** has the placement space **B12** formed at each of the two ends thereof in the length direction. The window air conditioner **2000** further includes the sealing assembly **B400**. The sealing assembly **B400** includes a fixation member **B401** and a sealing member **B402**. The fixation member **B401** is connected to the placement space **B12**. The sealing member **B402** is connected to the fixation member **B401** and adapted to be sealingly arranged between the window sash and the inner wall of the window.

(129) It should be understood that the sealing member **B402** is connected into the placement space **B12** by the fixation member **B401**. In the state of the window sash closing the window, the sealing

member **B402** is brought into contact with the window sash at one side thereof and with the inner wall of the window at the other side thereof to seal the window by the sealing member **B402**. Therefore, on the one hand, the sealing performance of the sealing assembly **B400** is improved, and on the other hand, the sealing assembly **B400** has a good sound insulation effect.

(130) Optionally, the sealing assembly **B400** corresponds to the placement space **B12** one-to-one. Therefore, the structure of the window air conditioner **2000** can be simplified and the sealing performance and sound insulation effect of the window air conditioner **2000** can be improved.

(131) Further, the sealing member **B402** may have an adjustable length which can be cut on site based on the distance between the side wall surfaces of the front housing **B200** and the rear housing **B300** and the inner wall surface of the window, such that the sealing member **B402** can provide better sealing to the window, thereby ensuring sealing of the window while the sealing assembly **B400** has a simpler structure. For example, the sealing member **B402** may be a sealing sponge.

(132) According to some embodiments of the present disclosure, the fixation member **B401** includes a rotation support **B4011** and a mounting member **B4012**. The rotation support **B4011** is fixed in the placement space **B12**. The mounting member **B4012** is rotatably arranged on the rotation support **B4011** in such a manner that the sealing assembly **B400** is capable of being rotated to be received within the accommodation groove **B600**. In this way, it is possible to only facilitate mounting of the fixation member **B401**, but also facilitate a rotation of the mounting member **B4012** relative to the rotation support **B4011**, such that the sealing assembly **B400** is easily stored to reduce a space occupied by the sealing assembly **B400**.

(133) Further, the mounting member **B4012** has a pivoting shaft **B4014** provided thereon, the rotation support **B4011** has a pivoting hole **B4015** formed therein, and the pivoting shaft **B4014** and the pivoting hole **B4015** are configured to be rotatably engaged with each other. In this manner, the pivoting shaft **B4014** and the pivoting hole **B4015** can be engaged with each other to facilitate a smooth rotation of the mounting member **B4012** and an improvement of the reliability of the rotation of the mounting member **B4012**.

(134) According to some embodiments of the present disclosure, the mounting member **B4012** has a mounting groove **B4013** formed therein, and a part of the sealing member **B402** is mounted in the mounting groove **B4013**. Therefore, by fixing the part of the sealing member **B402** in the mounting groove **B4013**, a connection structure between the sealing member **B402** and the mounting member **B4012** is simpler and more reliable.

(135) As illustrated in FIG. 24 to FIG. 26, in some embodiments of the present disclosure, the mounting groove **B4013** has a locking ridge **B4016** provided on each of opposite side surfaces thereof, and the locking ridge **B4016** is configured to be in contact with the sealing member **B402** to position the sealing member **B402** within the mounting groove **B4013**. In this manner, a strength of a connection between the sealing member **B402** and the mounting member **B4012** is improved, and the connection between the sealing member **B402** and the mounting member **B4012** is more stable and reliable.

(136) Some specific embodiments of a window air conditioner **3000** according to the embodiments of the present disclosure will be described below with reference to FIG. 28 to FIG. 42.

(137) As illustrated in FIG. 28, the window air conditioner **3000** is adapted to be mounted at a window in a wall body **C20** for cooling or heating an indoor environment.

(138) As illustrated in FIG. 28 to FIG. 30, the window air conditioner **3000** includes a base **C100**, a casing body **C200**, a support member **C500**, and a middle partition plate **C400**. The casing body **C200** has an accommodation groove **C230** formed therein. The accommodation groove **C230** divides the casing body **C200** into a front housing **C210** and a rear housing **C220**, and is adapted to allow a window sash **C30** located at the window in the wall body **C20** to extend therein. The support member **C500** is arranged on the base **C100** and corresponds to the accommodation groove **C230**. The middle partition plate **C400** is mounted on the support member **C500** and is adapted to allow the window sash **C30** to abut therewith. Here, the middle partition plate **C400** has a

positioning member **C440** formed on a bottom surface thereof, and the support member **C500** has a socket **C510** formed thereon. The socket **C510** corresponds to the positioning member **C440**. The middle partition plate **A400** is connected to the socket **C510** of the support member **C500** by inserting the positioning member **C440** with the socket **C510**.

(139) As illustrated in FIG. 34 and FIG. 35, during mounting the middle partition plate **C400**, since the support member **C500** is formed with the socket **C510** corresponding to the positioning member **C440**, the positioning member **C440** of the middle partition plate **C400** is aligned with the socket **C510** on the support member **C500** first, and then the middle partition plate **C400** is pressed from the top to bottom to insert the positioning member **C440** of the middle partition plate **C400** into the socket **C510** of the support member **C500**, thereby connecting and fixing the middle partition plate **C400** to the support member **C500**. That is, the middle partition plate **C400** is engaged with the support member **C500** by inserting the positioning member **C440** into the socket **C510**, and thus the middle partition plate **C400** can be detached or mounted by pulling the positioning member **C440** out of the socket **C510** or pressing the positioning member **C440** into the socket **C510**, which reduces the use of a screw structure or an engagement structure, thereby simplifying a mounting manner of the middle partition plate **C400** and realizing quick mounting of the middle partition plate **C400**. In addition, since such an insertion connection method does not require a large mounting space, an interference with a side wall of the accommodation groove **C230** during mounting can be reduced.

(140) After the window air conditioner **3000** is mounted at the window in the wall body **C20**, the window sash **C30** is pulled downwardly to extend into the accommodating groove **C230** (as illustrated in FIG. 28) of the window air conditioner **3000** until a lower edge of the window sash **C30** is brought into contact with and abuts with an upper surface of the middle partition plate **C400**. The window sash **C30** blocks a gap between the both sides of the window air conditioner **3000** and side walls of the window to avoid a leakage of indoor cold or heated air from the gap to the outdoor space. It should be noted that the window sash **C30** may also be replaced with a window louver, or a curtain or the like that blocks light, or other protective window plates that can block an external object from entering the indoor space.

(141) In the above process of mounting the window air conditioner **3000**, compared with a window sash in the related art that can only be pulled up to a top surface of the window air conditioner **3000**, the window sash **C30** according to this embodiment can be inserted into the window air conditioner **3000**. In this way, it is possible for the window sash **C30** to block a space between the both sides of the window air conditioner **3000** and a bottom wall of the window, thereby increasing an occupation area of the window sash **C30**.

(142) In the technical solution of the present disclosure, the positioning member **C440** is formed on the bottom surface of the middle partition plate **C400**, the support member **C500** is formed on a bottom portion of the casing body **C200**, the support member **C500** is formed with the socket **C510** corresponding to the positioning member **C440**, and the middle partition plate **C400** is inserted in and connected to the socket **C510** of the support member **C500** with the positioning member **C440**, such that the middle partition plate **C400** can be connected and fixed to the support member **C500**. It can be seen that in the window air conditioner **3000** of the present disclosure, the middle partition plate **C400** and the support member **C500** is engaged with each other by inserting the positioning member **C440** into the socket **C510**, which reduces the use of the screw structure or the engagement structure, thereby simplifying the mounting manner of the middle partition plate **C400**. Further, it is possible to lower a difficulty in mounting the middle partition plate **C400**, thereby realizing quick mounting of the middle partition plate **C400** and further improving the mounting efficiency.

(143) As illustrated in FIG. 31, in some embodiments, the window air conditioner **3000** further includes a water receiving tray **C600** mounted on the base **C100**. The water receiving tray **C600** corresponds to the front housing **C210**. On this basis, the support member **C500** may be formed on

the water receiving tray C600 to be arranged to the base C100 by the water receiving tray C600. Alternatively, the support member C500 may be formed directly on the base C100. The support member C500 may be a separately molded support part, or may be formed into one piece with the base C100, or formed into one piece with the water receiving tray C600.

(144) Here, specifically, the support member C500 and the water receiving tray C600 are integrally formed to reduce fabrication procedures. The water receiving tray 600 has a rear end extending to a position below the middle partition plate C400 and has the support member C500 at a part thereof located below the middle partition plate C400. In this manner, the support member C500 can support the middle partition plate C400, and water on the middle partition plate C400 can be directly discharged downwardly into the water receiving tray C600.

(145) As illustrated in FIG. 30 to FIG. 35, in some embodiments, the middle partition plate C400 includes a bottom plate C410 and end side plates C430 formed at both ends of the bottom plate C410. The positioning member C440 is formed on the bottom plate C410 of the middle partition plate C400. The number of the positioning members C440 on the middle partition plate C400 is not specifically limited herein. One positioning member C440 may be provided and formed at a middle position of the middle partition plate C400. Alternatively, a plurality of positioning members C440 may be provided and arranged at intervals in a length direction of the middle partition plate C400.

(146) Specifically, a plurality of positioning members C440 are provided and arranged at intervals in a length direction of the middle partition plate C400. Correspondingly, a plurality of sockets 510 are provided. The plurality of positioning members C440 is inserted in and connected to the plurality of sockets C510 in one-to-one correspondence. By connecting and inserting the middle partition plate C400 to the plurality of sockets C510 of the support member C500 with the plurality of positioning members C440, a fixing effect of a connection between the middle partition plate C400 and the support member 500 can be enhanced, and thus loosening of the middle partition plate C400 is less likely to occur.

(147) As illustrated in FIG. 35 to FIG. 37, further, the plurality of positioning members C440 includes a first positioning member C441 and a second positioning member C442 which are different from each other in shape. The plurality of sockets C510 includes a first socket C511 for an insertion of the first positioning member C441 and a second socket C512 for an insertion of the second positioning member C442.

(148) Since the first positioning member C441 and the second positioning member C442 have different shapes, the middle partition plate C400 can be pressed and inserted into the support member C500 only when the first positioning member C441 corresponds to the first socket C511 and the second positioning member C442 corresponds to the second socket C512 during mounting the middle partition plate A400. However, if the first positioning member C441 corresponds to the second socket C512 or another socket C510, and the second positioning member C442 corresponds to the first socket C511 or another socket 510, it is difficult to press and insert the middle partition plate C400 into the support member C500, thereby alerting an installation personnel that the middle partition plate C400 should be reversed. After the middle partition plate C400 is reversed, the first positioning member C441 can correspond to the first socket C511 and the second positioning member 442 can correspond to the second socket C512, such that the middle partition plate A400 can be pressed and inserted into the support member 500 without visually calibrating whether each positioning member C440 corresponds to a corresponding one of the plurality of sockets C510 by an user, thereby achieving a blind insertion.

(149) As illustrated in FIG. 37 and FIG. 38, the shape of each of the first positioning member C441 and the second positioning member C442 is not specifically limited herein. For example, the first positioning member C441 may have a circle, an oval, or a polygon shape (such as a triangular, square, trapezoidal, pentagonal shape or the like.), or other irregular shapes. The second positioning member C442 only needs to be different from the first positioning member C441. Specifically, the first positioning member C441 has a trapezoidal shape, and the second positioning member C442

has a square shape. Correspondingly, an insertion chamber of the first socket **C511** has the same shape as the first positioning member **C441** for an insertion of the first positioning member **C441**, and an insertion chamber of the second socket **C512** has the same shape as the second positioning member **C442** for an insertion of the second positioning member **C442**.

(150) As illustrated in FIG. 36 to FIG. 38, it is taken into consideration herein that during fabrication of the middle partition plate **C400** and the support member **C500**, due to possible errors in dimensions of individual structures, a distance between the first positioning member **C441** and the second positioning member **C442** may be different from a distance between the first socket **C511** and the second socket **C512**, which resulting in a difficulty in precisely aligning the middle partition plate **C400** with the support member **C500** easily.

(151) In view of this, a volume of the insertion chamber of the first socket **C511** is set to be greater than that of the insertion chamber of the first positioning member **C441**. During mounting the middle partition plate **C400**, when the first positioning member **C441** is inserted into the insertion chamber of the first socket **C511**, a certain space may be left in the insertion chamber, and a position of the partition plate in a leftward-rightward direction thereof can be adjusted due to such a left space to align the second positioning member **C442** with the second socket **C512**, such that the second positioning member **C442** can be inserted into the second socket **C512**. In this way, a mounting resistance caused by the dimensional errors can be overcome, thereby effectively lowering a difficulty in calibration.

(152) In some embodiments, the middle partition plate **C400** has a bisecting line that bisects its length, and the first positioning member **C441** and the second positioning member **C442** are asymmetrical about the bisecting line. During mounting the middle partition plate, it should be that the first positioning member **C441** is inserted into the first socket **C511** and the second positioning member **C442** is inserted into the second socket **C512**. If the middle partition plate is rotated by 180°, although the first positioning member **C441** is close to the second socket **C512**, the first positioning member **C441** is misaligned with the second socket **C512**, and similarly, the second positioning member **C442** is misaligned with the first socket **C511**, and thus an alignment is impossible, thereby alerting the installation personnel to reverse the middle partition plate **C400**. In this manner, the middle partition plate **C400** can be mounted in place without visually calibrating whether each positioning member **C440** corresponds to a corresponding one of the plurality of sockets **C510** by the user, thereby achieving the blind insertion.

(153) As illustrated in FIG. 30, FIG. 31, and FIG. 38, in some embodiments, the middle partition plate **C400** includes the bottom plate **C410** and end side plates **C430** formed at both ends of the bottom plate **C410**, and the bottom plate **C410** is formed with the positioning member **C440** on the bottom surface thereof to be inserted in and connected to the socket **C510** of the support member **C500**. In addition, the two end side plates **C430** are connected to opposite side walls of the base **C100**, respectively, to improve mounting stability of the middle partition plate **C400**.

(154) Optionally, one of the side wall of the base **C100** and the end side plates **C430** of the middle partition plate **C400** has a laterally protruding snap buckle **C434** provided thereon, and the other of the side wall of the base **C100** and the end side plates **C430** of the middle partition plate **C400** has an engagement hole **C110** formed therein. The engagement hole **C110** corresponds to the snap buckle **C434** and is adapted to be engaged with the snap buckle **C434**. The mounting of the middle partition plate **C400** will be described below by taking the snap buckle **C434** being arranged on the end side plate **C430** and the engagement hole **C110** being arranged on the side wall of the base **C100** as an example.

(155) During mounting the middle partition plate **C400**, the end side plates **C430** at both ends of the middle partition plate **C400** are pressed inwardly in such a manner that each end side plate **C430** of the middle partition plate **C400** extends into an inner side of the side wall of the base **C100** corresponding to the end side plate **C430**. Then, the positioning member **C440** of the middle partition plate **A400** is inserted into the socket **C510** of the support member **C500** from top to

bottom. In this process, each of the end side plates **C430** at both ends of the middle partition plate **A400** is compressed inwardly against the side wall of the base **C100** until the snap buckle **C434** on the end side plate **C430** is brought into contact with the engagement hole **C110** on the side wall of the base **C100** in response to the positioning member **C440** of the middle partition plate **C400** being inserted into the socket **C510** of the support member **C500** in place. In this case, a compress force of the side wall of the base **C100** against the end side plate **C430** is removed, and the end side plate **C430** is elastically biased back to its original position to urge the snap buckle **C434** into the engagement hole **C110** in the side wall of the base **C100**, thereby connecting the end side plate **C430** to the side wall of the base **C100**.

(156) As illustrated in FIG. 38 and FIG. 39, optionally, each of the end side plates **C430** includes a first plate body **C431** connected to the bottom plate **C410** and a second plate body **C432** connected to the first plate body **C431**. Here, the second plate body **C432** is formed with the snap buckle **C434**, and the second plate body **C432** and the first plate body **C431** are connected to each other at a connection platform **C433**. When the snap buckle **C434** is engaged with the engagement hole **C110**, the connection platform **C433** abuts with an upper edge of the side wall of the base **C100** (as illustrated in FIG. 30 and FIG. 31). In this manner, an end portion of the middle partition plate **C400** can be supported by the side wall of the base **C100**, and the support member **C500** is supported between both end portions of the middle partition plate **A400**, thereby improving the mounting stability of the middle partition plate **C400**.

(157) As illustrated in FIG. 30, FIG. 31, and FIG. 37, in some embodiments, the middle partition plate **C400** further includes side plates **C420** formed at opposite sides of the bottom plate **C410**. The two side plates **C420** are attached to a back plate of the front housing **C210** and a back plate of the rear housing **C220**, respectively. It is taken into consideration herein that, due to a possible gap between the side plate **C420** and the back plate of the front housing **C210**, enteral water may fall into an interior of the front housing **C210** through the gap.

(158) In view of this, to avoid such a situation, one of the side plates **C420** has a support platform **C421** formed on an inner surface thereof, and the support platform **C421** extends in a length direction of the side plate. The front housing **C210** has a notch **C211** formed at the back plate thereof, and the notch **C211** corresponds to the support platform **C421** and abuts with the support platform **C421** at an upper edge thereof. That is, the back plate of the front housing **C210** rests on the support platform **C421** of the side plate **C420** of the middle partition plate **C400** to reduce the gap between the back plate of the front housing **C210** and the side plate **C420** of the middle partition plate **C400**. The external water is guided by the back plate of the front housing **C210** to flow towards the side plates **C420** of the middle partition plate **C400**, and then flows into an interior of the middle partition plate **C400** from the side plate **C420**.

(159) As illustrated in FIG. 38 and FIG. 40, since the back plate of the front housing **C210** rests on the support platform **C421** of the side plate **C420** of the middle partition plate **C400**, a part of a weight of the front housing **C210** is applied to the side plate **C420** of the middle partition plate **C400**. To this end, the side plate **C420** has a plurality of reinforcing ribs **C422** formed at a position on an outer surface thereof corresponding to the support platform **C421**. The plurality of reinforcing ribs **C422** is arranged at intervals in an extension direction of the support platform **C421**. A strength of the side plate **C420** of the middle partition plate **C400** can be strengthened by the plurality of reinforcing ribs **C422** to prevent the side plate **C420** from being bent and deformed by the back plate of the front housing **C210**.

(160) As illustrated in FIG. 32, FIG. 33, and FIG. 35 to FIG. 37, based on any one of the above embodiments, the middle partition plate **C400** further has a fixation part **C450** formed at a middle portion thereof, the support member **C500** has an engagement member **C520** formed thereon. The engagement member **C520** corresponds to the fixation part **C450**, and the fixation part **C450** is connected and fixed to the engagement member **C520** by a connection member. After the middle partition plate **C400** is inserted in and connected to the support member **C500**, the fixation part

C450 of the middle partition plate **C400** and the engagement member **C520** of the support member **C500** are connected to each other by the connection member to further secure the middle partition plate **C400**.

(161) Specifically, the bottom plate **C410** of the middle partition plate **A400** is recessed downwardly in a middle portion thereof to form a recess. The recess is formed with the fixation part **C450** at a bottom portion thereof. The fixation part **C450** has a fixation hole formed therein. The engagement member **C520** has an engagement hole **C521** formed therein. The engagement hole **C521** corresponds to the fixation hole, and is adapted to be connected to the fixation hole on the fixation part **C450** by the connection member. The connection member may be a screw or a pin.

(162) With continued reference to FIG. 32, FIG. 33, and FIG. 37, further, the casing has a fixation plate **C1211** extending therefrom towards the fixation part **C450**, and thus the casing body **C200** has the fixation plate **C1211** extending therefrom towards the fixation part **C450**. The connection member passes through the fixation plate **C1221**, the fixation part **C450** and the engagement member **C520** in sequence to integrally connect and fix the middle partition plate **C400** to the casing body **C200** and the support member **C500**, thereby strengthening a fixing strength of the middle partition plate **C400**, such that the middle partition plate **C400** is less liable to loosening.

(163) Specifically, a housing **C120** of the casing body **C200** is divided into the front housing **C210** and the rear housing **C220** by the accommodation groove **C230**, and the back plate of the front housing **C210** and the back plate of the rear housing **C220** is form as two side walls of the accommodation groove **C230**, respectively. Therefore, the fixation plate **C1221** can be formed on the back plate of the front housing **C210** or on the back plate of the rear housing **C220**. Here, specifically, the fixation plate **C1221** is formed on the back plate of the rear housing **C220**.

(164) As illustrated in FIG. 41, based on any one of the above embodiments, the window air conditioner **3000** further includes a compressor, an outdoor heat exchanger, an outdoor fan, an indoor heat exchanger, and an indoor fan. Here, the compressor, the outdoor heat exchanger, and the outdoor fan are mounted within the rear housing **C220**, and the indoor heat exchanger and the indoor fan are mounted within the front housing **C210**, such that noises generated by members at an outdoor side can be prevented from propagating to an indoor side, thereby achieving noise reduction effect.

(165) As illustrated in FIG. 41 and FIG. 42, in some embodiments, the window air conditioner **3000** further includes a sealing assembly **C300**. The sealing assembly **C300** is movably mounted on the middle partition plate **C400**, and is switchable between a storage state and an operation state through movement. Here, in the storage state, the sealing assembly **C300** is received in the accommodation groove **C230**, and in the operation state, the sealing assembly **C300** extends laterally from the accommodation groove **C230** and is adapted to be abutted against the window sash **C30** and/or an inner wall of the window.

(166) After the window air conditioner **3000** is mounted at the window, the sealing assembly **C300** is moved to the operation state in such a manner that the sealing assembly **300** extends laterally from the accommodation groove **C230** of the window air conditioner **3000** with a bottom surface of the sealing assembly **C300** abutted against the bottom wall of the window. Then, the window sash **C30** is pulled downwardly to extend into the accommodation groove **C230** of the window air conditioner **3000** until the lower edge of the window sash **C30** is brought into contact and abuts with the middle partition plate **C400** and the sealing assembly **C300**. The sealing assembly **C300** is filled in a gap between the window sash **C30** and the bottom wall of the window and seals the gap to reduce an amount of cold or heated air that leaks from the indoor space to the outdoor space through the gap. When the window air conditioner **3000** is not needed or is transported, the sealing assembly **C300** is moved to the storage state to reduce a space occupied by the sealing assembly **C300**, thereby facilitating storage or packaging of the window air conditioner **3000**.

(167) Movable mounting for the sealing assembly **C300** may be implemented in various ways. For example, the sealing assembly **C300** may be slidably mounted on the middle partition plate **C400**.

Alternatively, the sealing assembly C300 may be rotatably mounted on the middle partition plate C400. Additionally, alternatively, the sealing assembly C300 may be mounted on the middle partition plate C400 in a resiliently retractable manner. Specifically, the sealing assembly C300 is rotatably connected to the middle partition plate C400 to allow the sealing assembly C300 to be switchable between the operation state and the storage state through a rotation.

(168) As illustrated in FIG. 34, FIG. 37, and FIG. 41, further, the middle partition plate C400 further has a placement space C460 formed therein for mounting the sealing assembly C300. A side wall of the placement space C460 is spaced apart from the end side plate C430 of the middle partition plate C400 to form a clearance groove C470 between the side wall of the placement space C460 and the end side plate C430. The clearance groove C470 can provide greater elasticity to the end side plate C430. The end side plate C430 can be elastically biased towards an inner side of the clearance groove C470, thereby allow the end side plate C430 to be biased towards the inner side of the clearance groove C470 by compression of the side wall of the base C100 during mounting the middle partition plate C400, until the snap buckle C434 on the end side plate C430 is brought into contact with the engagement hole on the side wall of the base C100. When the snap buckle C434 on the end side plate C430 is in contact with the engagement hole on the side wall of the base C100, the end side plate C430 is elastically biased back to its original position to urge the snap buckle C434 into the engagement hole, thereby connecting and fixing the end side plate C430 to the side wall of the base C100.

(169) In the description of this specification, descriptions with reference to the terms “an embodiment,” “some embodiments,” “illustrative embodiments,” “an example,” “a specific example,” “some examples,” etc., are intended to mean that specific features, structure, materials, or characteristics described in conjunction with the embodiment or example are included in at least one embodiment or example of the present disclosure. In this specification, the schematic representations of the above terms do not necessarily refer to the same embodiment or example. Moreover, the described specific features, structures, materials or characteristics may be combined in any one or more embodiments or examples in a suitable manner.

(170) Although embodiments of the present disclosure have been illustrated and described, it should be understood for those skilled in the art that various changes, modifications, replacements, and variations can be made to these embodiments without departing from the principles and spirit of the present disclosure. The scope of the invention shall be defined by the claims as appended and their equivalents.

Claims

1. A sealing assembly comprising: a first member configured to be connected to a casing of a window air conditioner to be supported at a window, the casing having an accommodation groove, and the first member including a pivoting shaft; a sealing member connected to the first member and configured to be sealingly arranged between: a window sash of the window that is at least partially extended in the accommodation groove, and an inner wall of the window; and a second member configured to be fixed at the casing, the first member being rotatably disposed on the second member to enable the sealing assembly to rotate to be received in the accommodation groove, and the second member including a pivoting hole configured to be rotatably engaged with the pivoting shaft of the first member.

2. The sealing assembly according to claim 1, wherein the first member has a mounting groove, a part of the sealing member being mounted in the mounting groove.

3. The sealing assembly according to claim 2, wherein the mounting groove includes a locking ridge provided on each of two opposite side surfaces of the mounting groove and configured to be in contact with the sealing member to position the sealing member within the mounting groove.

4. The sealing assembly according to claim 1, wherein the second member includes a rib provided

on each of two opposite side walls of the second member and configured to be in contact with an inner wall surface of the accommodation groove at a corresponding side.

5. The sealing assembly according to claim 4, wherein the rib includes a guide chamfer provided at a lower end of the rib.

6. The sealing assembly according to claim 4, wherein the second member further includes a sealing plate provided on each of the two opposite side walls of the second member, the sealing plate extending towards the inner wall surface of the accommodation groove at the corresponding side and being located above the rib on the corresponding side.

7. The sealing assembly according to claim 1, further comprising: an angle positioning assembly configured to cooperate with the second member and the first member to position the first member at a predetermined angle when the first member rotates to the predetermined angle.

8. A window air conditioner comprising: a casing having an accommodation groove configured to receive at least a part of a window sash of a window; and a sealing assembly configured to be sealingly arranged between the window sash and an inner wall of the window, and including: a first member configured to be connected to the casing and including a pivoting shaft; a sealing member connected to the first member and configured to be sealingly arranged between: the window sash that is at least partially extended in the accommodation groove, and the inner wall of the window; and a second member configured to be fixed at the casing, the first member being rotatably disposed on the second member to enable the sealing assembly to rotate to be received in the accommodation groove, and the second member including a pivoting hole configured to be rotatably engaged with the pivoting shaft of the first member.

9. The window air conditioner according to claim 8, wherein the casing includes: a base; a rear housing fixed at the base and configured to accommodate an outdoor heat exchanger; a front housing fixed at the base and arranged to be spaced apart from the rear housing to form the accommodation groove; and a middle partition plate fixed at the base and located within the accommodation groove, a front end and a rear end of the middle partition plate being engaged with the rear housing and the front housing, respectively.

10. The window air conditioner according to claim 9, further comprising: a water receiving tray mounted on the base and including drainage groove configured to drain water to the base; wherein: the middle partition plate includes a bottom plate and side plates formed at two opposite sides of the bottom plate, a receiving groove being formed by the side plates and the bottom plate; the receiving groove is recessed downwardly at each of two ends of the receiving groove to form a placement space; a first drainage hole is formed at one end of the placement space, the first drainage hole being located above the drainage groove of the water receiving tray and configured to drain water to the water receiving tray; and a second drainage hole is formed at another end of the placement space and is configured to drain water to the base directly.

11. The window air conditioner according to claim 10, wherein the water receiving tray partially extends to a position below the middle partition plate and includes a support member located below the middle partition plate, the support member supporting the middle partition plate.

12. The window air conditioner according to claim 10, wherein a water overflowing hole is formed in one side plate of the side plates that is closest to the rear housing among the side plates, the water overflowing hole penetrating the one side plate and being configured to drain water to a part of the base corresponding to an outdoor side.

13. The window air conditioner according to claim 9, wherein: the base includes a support member, the support member including a socket; and the middle partition plate includes a positioning member provided on a bottom surface of the middle partition plate, the positioning member being configured to be inserted in the socket.

14. The window air conditioner according to claim 13, wherein: the positioning member is one of a plurality of positioning members provided at the bottom surface of the middle partition plate, the plurality of positioning members being arranged at intervals in a length direction of the middle

partition plate and including a first positioning member and a second positioning member having different shapes; and the socket is one of a plurality of sockets of the support member, the plurality of sockets including a first socket configured to receive the first positioning member and a second socket configured to receive the second positioning member.

15. The window air conditioner according to claim 13, wherein: the middle partition plate includes a bottom plate and end side plates formed at two ends of the bottom plate; the positioning member is formed at a bottom surface of the bottom plate; and the end side plates are connected to two opposite side walls of the base, respectively.

16. The window air conditioner according to claim 15, wherein each of the end side plates includes a first plate body connected to the bottom plate and a second plate body connected to the first plate body, a connection platform being formed at a location where the second plate body and the first plate body are connected to each other, and the connection platform abutting against an upper edge of the corresponding side wall of the base.

17. The window air conditioner according to claim 15, wherein: the middle partition plate further includes side plates formed at two opposite sides of the bottom plate, a support platform being formed at an inner surface of one side plate of the side plates and extending in a length direction of the one side plate; and a notch is formed at a back plate of the front housing, an upper edge of the notch abutting against the support platform.
